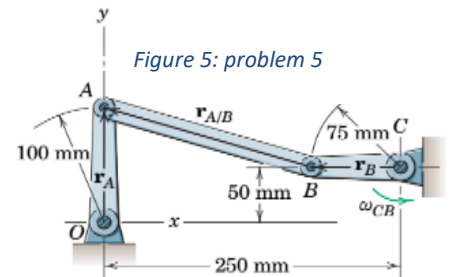


Figure 4: PROBLEM 4

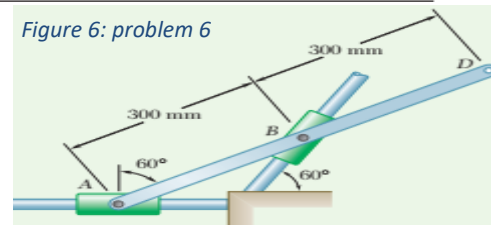
β is 15° , determine (a) the angular velocity of bar AB , (b) the velocity of the center of bar AB .

ACCELERATION ANALYSIS

5. Crank CB has a constant counterclockwise angular velocity of 2 rad/s in the position shown during a short interval of its motion. Determine the angular acceleration of links AB and OA for this position.

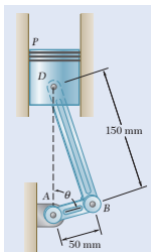
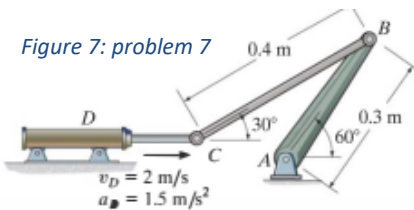


6. Collars A and B are pin-connected to bar ABD and can slide along fixed rods. Knowing that, at the instant shown, the velocity of A is a constant 0.9 m/s to the right, determine the angular acceleration of AB and the acceleration of B .



7. The hydraulic cylinder is extending with the velocity and acceleration shown. Determine the angular acceleration of crank AB and link BC at the instant shown.

8. Knowing that crank AB rotates about point A with a constant angular velocity of 900 rpm clockwise, determine the acceleration of the piston P when θ is 60° .



assignment 1 (neatness has a better mark)

1. 3 and 6 for odd numbered id
2. 4 and 8 for even numbered id

Figure 8: problem8